PulpoCon 2025

Unknown unknowns with Continuous Profiling

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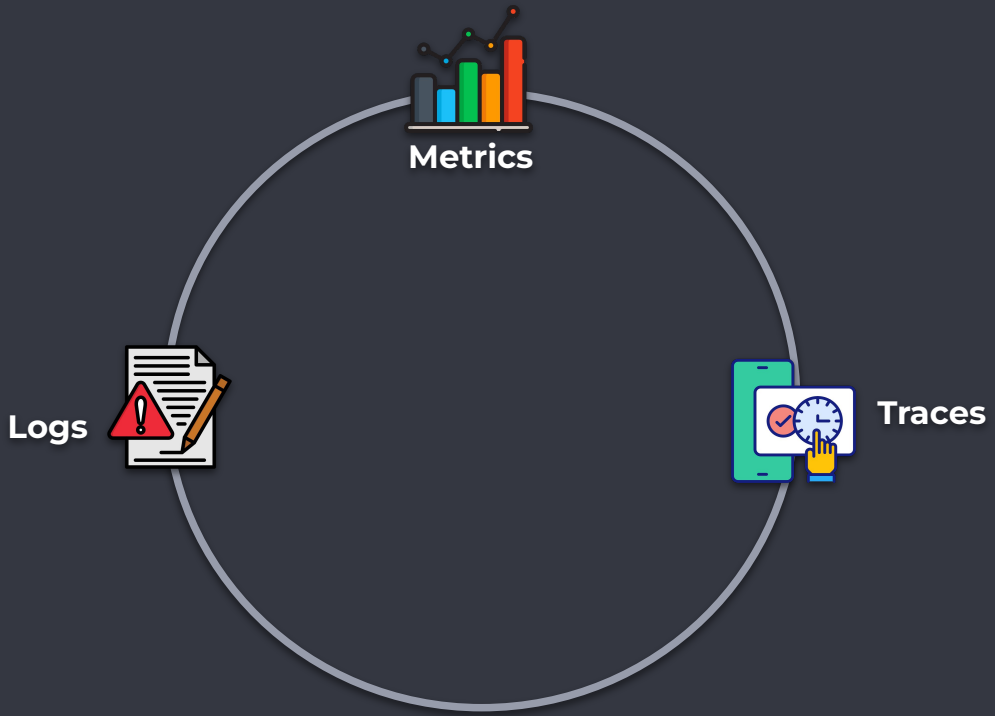
[Draft] Agenda

1. Observability pillars
2. Profiles specification
3. Profiling Agent architecture
4. Stack unwinding
5. Supported languages
6. Symbolization
7. Demo: Detecting unknown-unknowns of the OpenTelemetry Demo with Devfiler and the OpenTelemetry eBPF Profiler

The Three pillars of Observability



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What's happening in our systems?

{ http.error.rate = 75%, timestamp = ... }

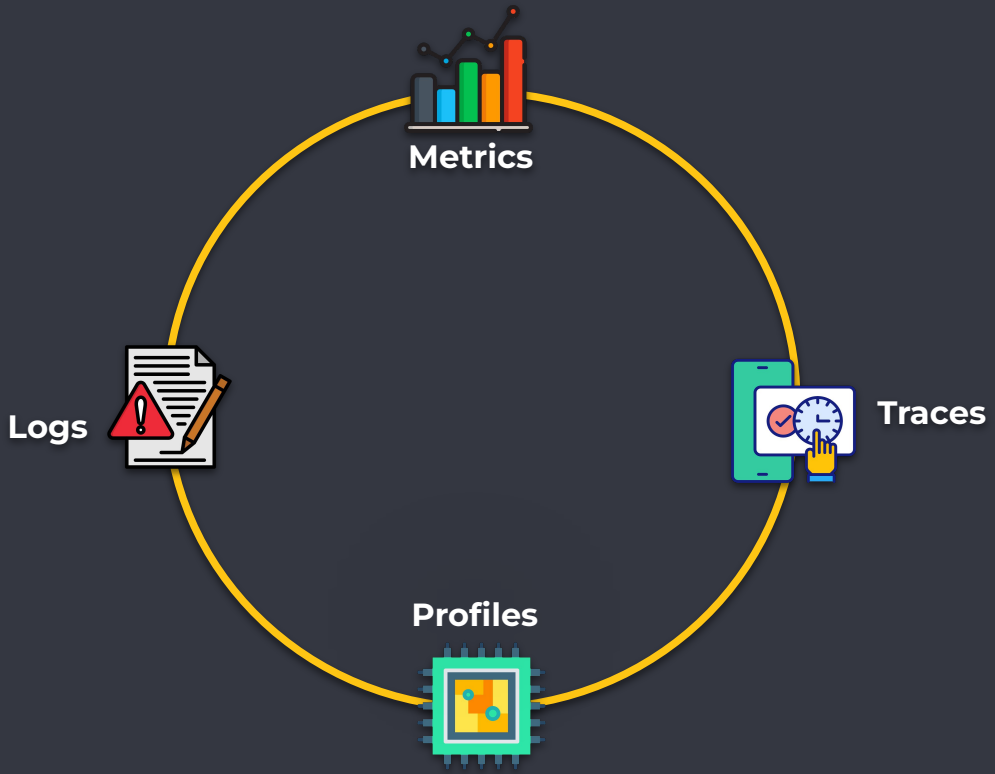
Why is it happening?

{ message = “user not found”, timestamp = ... }

Where is it happening?

{ name = “get_user”, span_id = ..., parent_id = ... }

The ~~Three~~ Four pillars of Observability



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Understand CPU usage quickly and completely

Unknown unknowns

- Why is the kernel on-CPU so much? What code-paths?
- Are the CPUs stalled on memory I/O?
- Which code-paths are allocating memory, and how much?
- Is a certain kernel function being called, and how often?
- What reasons are threads leaving the CPU?
- Regressions?

OpenTelemetry Profiles

```
dictionary:
  attributeTable:
    - key: process.executable.build_id.htlhash
      value:
        stringValue: 600DCAFE4A110000F2BF38C493F5FB92
    - key: profile.frame.type
      value:
        stringValue: native
    - key: host.id
      value:
        stringValue: localhost
  locationTable:
    - address: "111"
      attributeIndices:
        - 1
      mappingIndex: 0
  mappingTable:
    - attributeIndices:
        - 0
  stringTable:
    - samples
    - count
    - cpu
    - nanoseconds
resourceProfiles:
  - resource: {}
    scopeProfiles:
      - profiles:
          - attributeIndices:
              - 2
            periodType:
              typeStrindex: 2
              unitStrindex: 3
            sample:
              - locationsLength: 1
                timestampsUnixNano:
                  - "0"
            sampleType:
              - unitStrindex: 1
        scope: {}
```

OpenTelemetry announces support for profiling

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To Other Signals

```
profile:
  link:
    trace_id: 12345
    span_id: 67890
   ...
```

From Other Signals

```
log:
  profile_id: 88888
  trace_id: 12345,
  span_id: 67890,
  severity_text: "DEBUG",
  body: "Testing log",
   ...
```

OpenTelemetry Profiling Agent

Frictionless deployment, no code changes or app restarts, powered by eBPF.

Whole-system visibility: from the Kernel through user space.

Multiple High Level Language runtimes.

Extremely Low Overhead: aiming for: < 1% system CPU, ~250MB of RAM

Elastic donates profiling agent to OpenTelemetry

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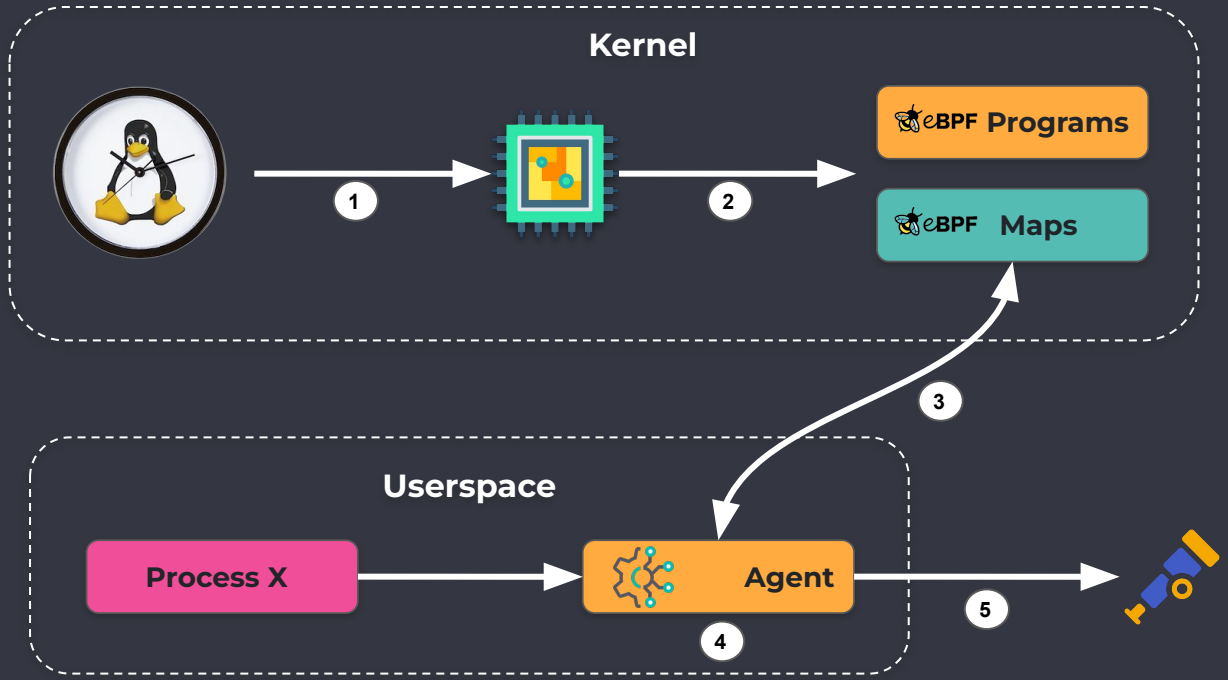
Profiling Agent Architecture



1. CPU interrupt (19Hz)
2. Stack unwinding
3. Process-specific data
4. Symbolization
5. OTLP Profiles

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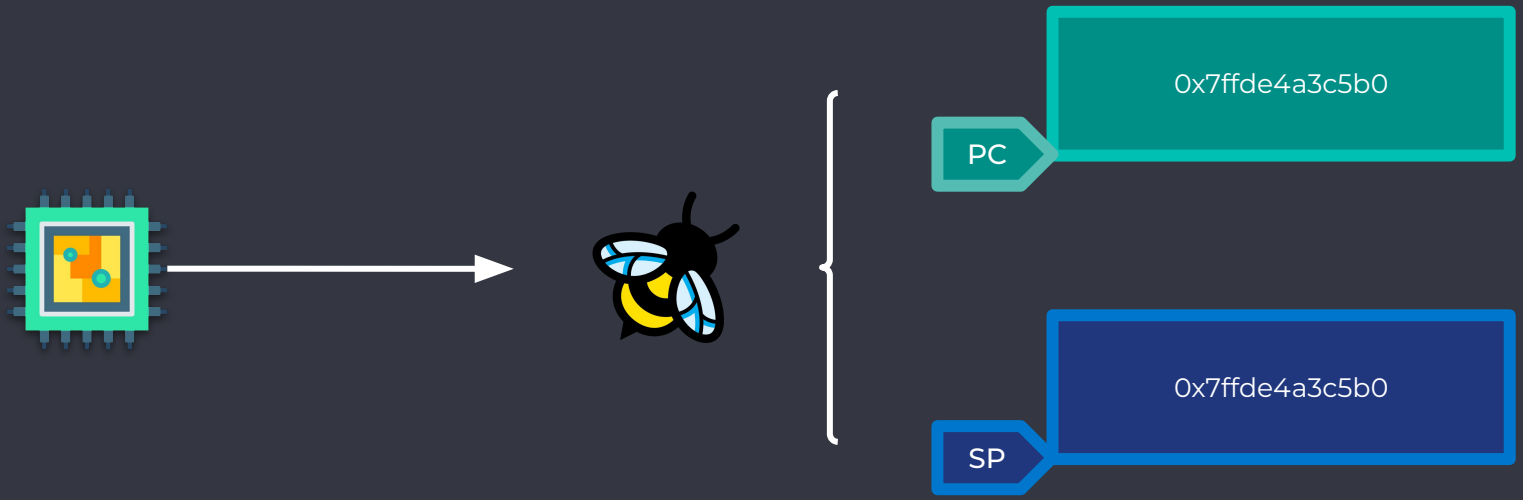
Stack unwinding (FPO)



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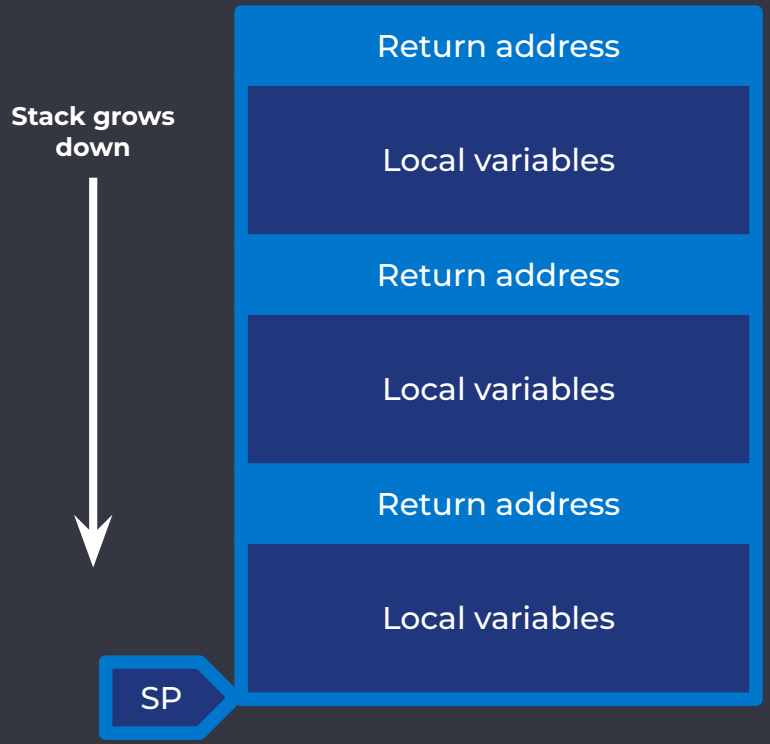
Stack unwinding (FPO)



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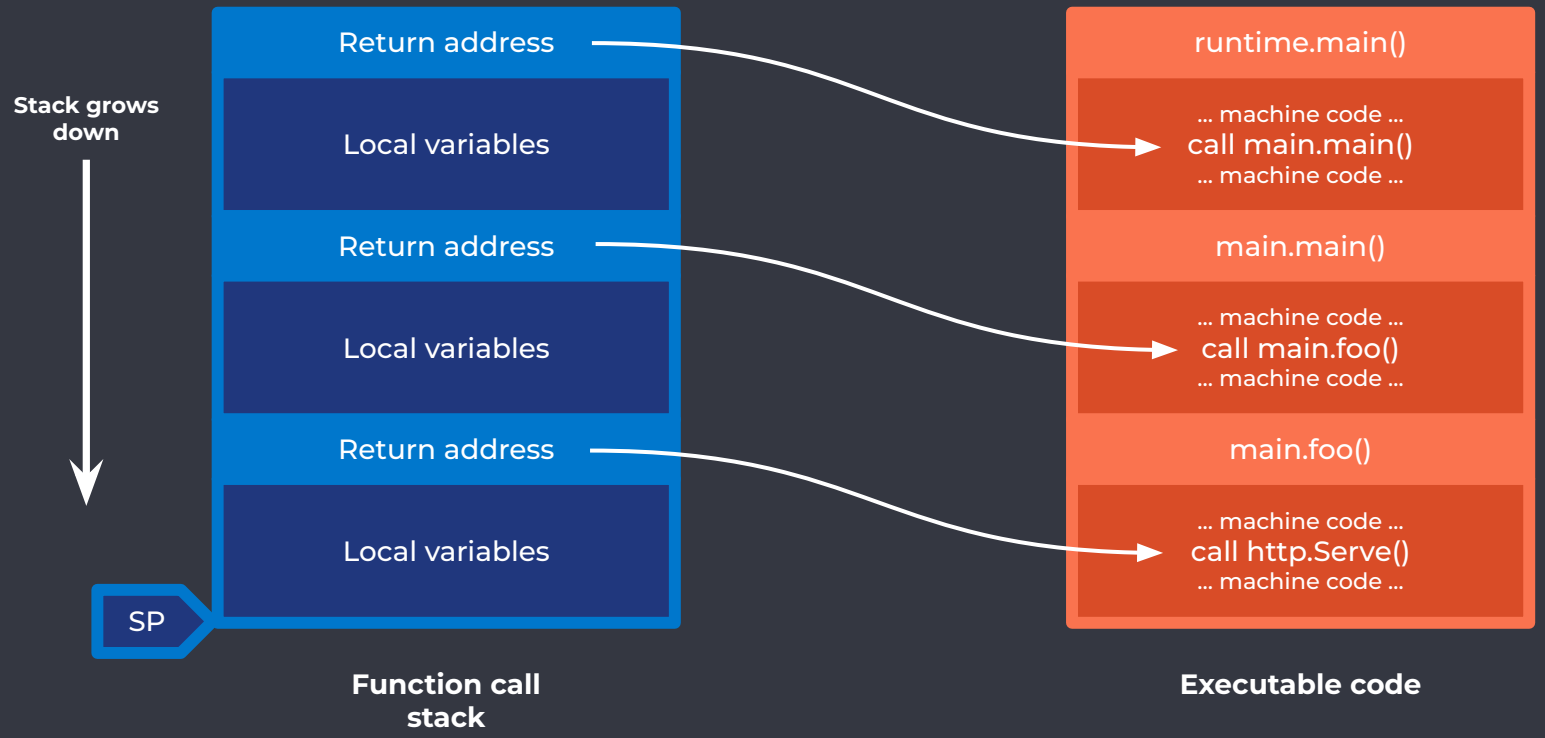
Stack unwinding (FPO)



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Stack unwinding

Native (C, C++, Go, ...) vs High Level Languages (Java, Python, ...)

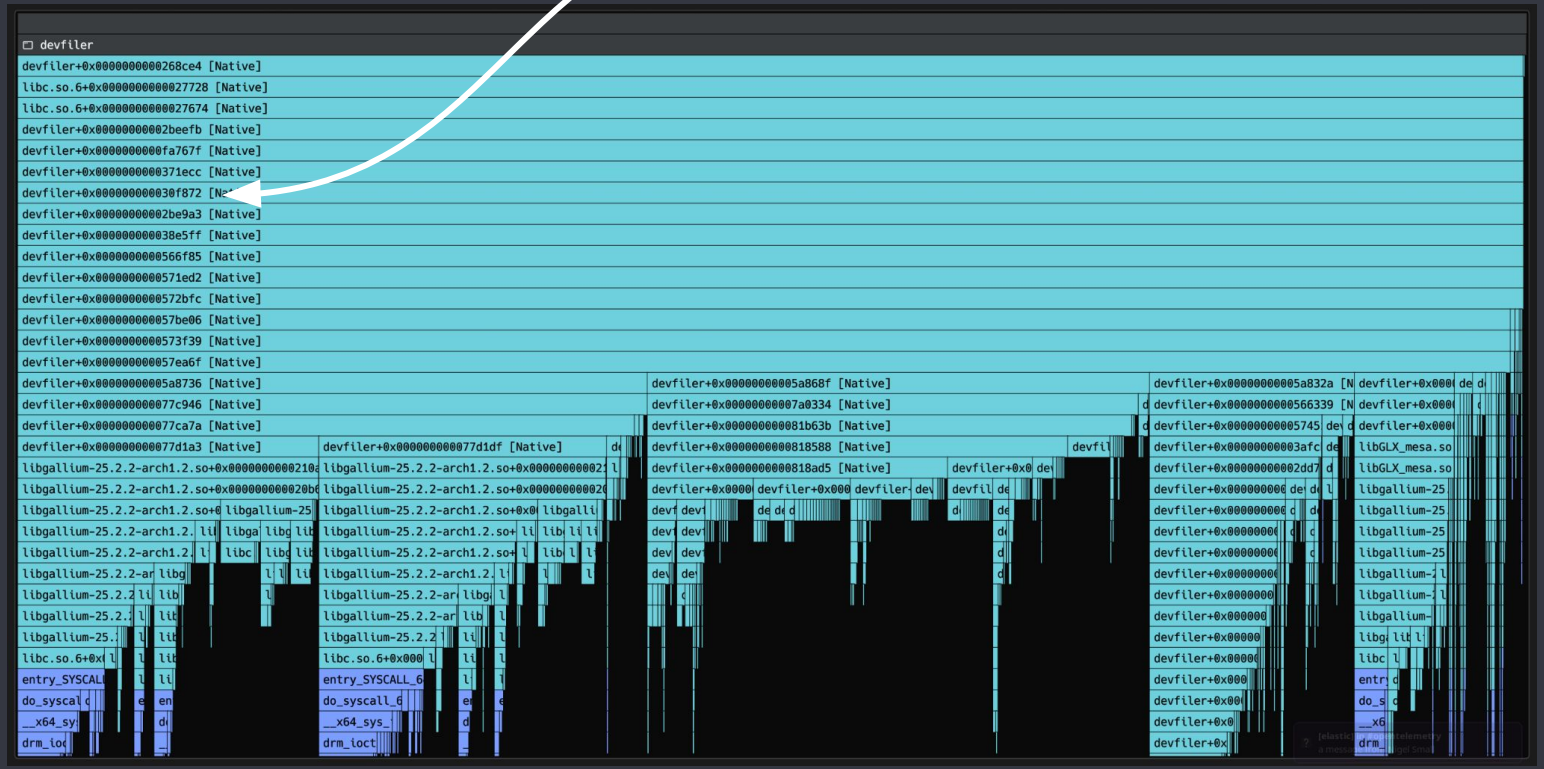
Native executables: via exception handling and unwinding mechanism (.eh_frame) on ELFs metadata for C++, Rust, C, etc. For Go, unwinding relies on the binary *.gopclntab* section.

```
String dump of section '.gopclntab':
  ...
  [19140f]  github.com/collector/receiver/xreceiver.NewFactory
  [19144b]  github.com/collector/receiver.NewFactory
  [19147d]  github.com/collector/receiver/xreceiver.factory.CreateDefaultConfig
```

HLL runtimes: Every language needs to keep track of a call stack.

- Reverse engineer how each HLL runtime does it.
- Prepare eBPF maps with struct offsets and other information to enable unwinding.
- Supporting new versions can be a matter of only updating a few offsets

foo() or bar() ? 🤔



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Symbolization

Stack trace offsets

```
moduleID1+0×6c8
moduleID1+0×9c9
moduleID2+0×c8714
moduleID2+0×281e1
```

Human-readable symbols

```
vmlinux: __x64_sys_nanosleep
vmlinux: do_syscall_64
pf-host-agent: runtime.usleep
pf-host-agent: runtime.runqgrab
```

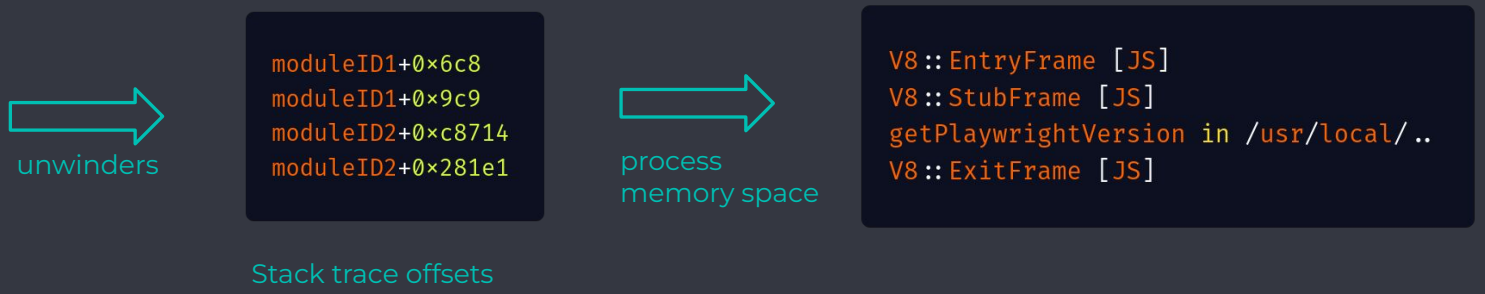
Replace addresses (offsets) with symbol names (also source filenames, line numbers when available)

Symbolization

Native vs High Level Languages

Native executables: may be symbolized on-target (if symbols always present) or post-collection (e.g. at the backend)

HLL runtimes: always present in target process memory space. The agent can perform symbolization on-target.



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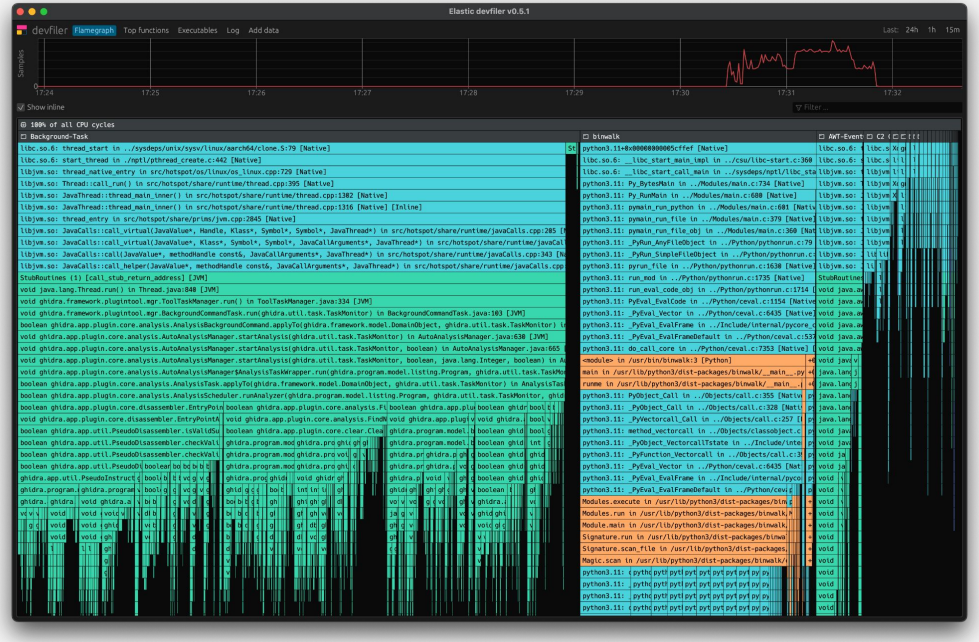
Supported languages

*DotNet is not supported on ARM64

Backends



- Any service that supports OTLP Profiling signal (Elastic, Polar Signals, Pyroscope, ..)
- Elastic devfiler: a standalone self-contained backend as a desktop application
- OpenTelemetry Collector receiver
- OpenTelemetry Collector eBPF Profiling Distribution (Experimental)

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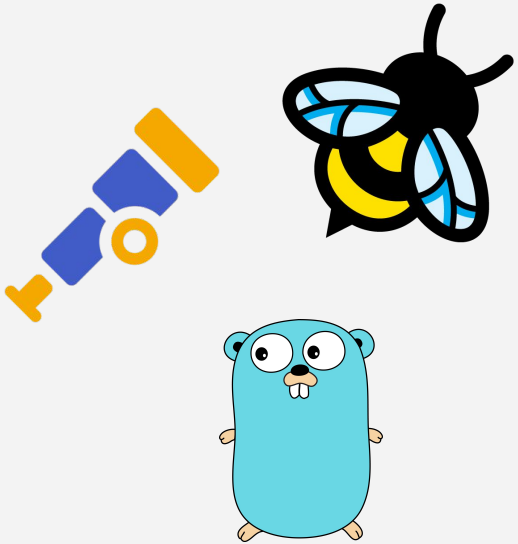
Demo



- DevFiler + Go (on-target) and Rust (post-collection) service's symbolization
- OpenTelemetry Astronomy Shop Demo deployments

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¡Gracias!


opentelemetry-ebpf-profiler


opentelemetry-proto


elastic/devfiler

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